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Original article

Unveiling the hidden burden of malnutrition in tuberculosis: A multifaceted evaluation using malnutrition universal screening tool and patient-generated subjective global assessment

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ABSTRACT

Problem considered: Tuberculosis (TB) remains a significant public health burden, especially in resource-limited countries. Malnutrition is a significant concern among TB patients, exacerbating symptoms and impeding recovery. Despite this, standardised nutritional screening remains underutilised in TB care programs. Accurate and routine nutritional assessment is essential to guide timely interventions and improve treatment outcomes among TB patients. This study aimed to assess malnutrition risk in TB patients using MUST and PG-SGA, evaluate disagreement between screening tools, and determine the limitations of Body Mass Index (BMI) based assessments.

Methods: A cross-sectional study was conducted among 194 adult TB patients receiving anti-TB therapy in Bihar, India. Nutritional risk was evaluated using MUST and PG-SGA, along with demographic, clinical, and anthropometric data. Statistical analyses included ANOVA, Bonferroni post hoc tests, chi-square tests, and Cohen's kappa to assess agreement between tools.

Results: The study revealed a high prevalence of malnutrition risk, with MUST detecting 74.2 % and PG-SGA detecting 88.1 % of the overall population. Underweight patients were disproportionately affected, with MUST identifying 96.5 % and PG-SGA identifying 88.3 % as high-risk. This study emphasises the limitations of BMI alone in assessing malnutrition risk and reveals near-random agreement ($\kappa = 0.010$, $p < 0.001$), suggesting BMI alone inadequately captures malnutrition risk in TB patients.

Conclusion: This study highlights a critical gap in malnutrition screening for TB patients, with MUST and PG-SGA detecting high but discordant risk levels. Given PG-SGA's higher sensitivity, policymakers in resource-limited settings should consider integrating it into TB programs for early nutritional intervention, while further research validates optimal screening approaches.

1. Introduction

Tuberculosis (TB) remains a leading cause of morbidity and mortality in resource-limited settings, where malnutrition, a key modifiable risk factor is often overlooked in clinical management.[1] Adequate nutrition is crucial for effective TB treatment, as optimal nutrient intake enhances medication efficacy and supports recovery. The relationship between TB and malnutrition is complex and bidirectional.[2] TB contributes to malnutrition by reducing appetite, causing malabsorption, increasing energy expenditure, and triggering systemic inflammation, which can lead to muscle wasting and nutrient deficiencies.[3] On the other hand, malnutrition can impair immune function, making patients more susceptible to TB infection, and reduce the effectiveness of TB treatment, leading to longer treatment durations or treatment failure.[4] Additionally, malnutrition can increase the risk of TB-related complications, such as respiratory failure or secondary infections, and

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impede healing, making patients more prone to TB-related complications like pleural effusions or lymphadenitis. This bidirectional relationship significantly compromises patient outcomes and exacerbates TB-related morbidity and mortality.[5]

Accurate assessment of malnutrition is crucial to guide interventions and mitigate its harmful effects. Moreover, it contributes to the cycle of poverty, disability, and disease burden, especially in vulnerable populations. The association between nutritional status and tuberculosis outcomes is well recognized, but routine nutritional assessment is still not widely adopted in many low- and middle-income countries (LMICs). The MUST and PG-SGA are widely used instruments for evaluating malnutrition risk.[6,7]

Previous studies have highlighted the impact of nutritional status on TB treatment outcomes; however, many rely solely on BMI or fail to use validated tools for comprehensive malnutrition assessment. While digital interventions, such as mobile applications, have shown promise in improving medication adherence among TB patients, as highlighted in recent systematic review, their integration with nutritional support strategies remains largely unexplored.[8] Moreover, existing studies often lack comparative evaluations between different nutritional screening methods, particularly in low-resource, high-TB-burden areas, such as Bihar, India.

This study was undertaken to address the critical knowledge gap in effective malnutrition screening among TB patients. Specifically, it aims to (1) assess the risk of malnutrition in TB patients using MUST and PG-SGA, (2) evaluate disagreement between screening tools, and (3) determine the limitations of BMI-based assessments. The findings are expected to inform public health strategies by identifying the most sensitive and appropriate malnutrition screening approach for integration into TB management programs in resource-constrained settings.

2. Methods

2.1. Study design

This cross-sectional study was conducted at the outpatient department of the ICMR-Rajendra Memorial Research Institute of Medical Sciences (RMRIMS), Patna, Bihar, India, from April 2024 to March 2025.

This site was selected for the research because it is a recognized regional centre for tuberculosis diagnosis and treatment, located in a high-burden area, and receives a large number of TB patients from diverse backgrounds.

2.2. Study population

2.2.1. Inclusion criteria

Individuals aged 18 years or older diagnosed with active TB (pulmonary or extrapulmonary) and receiving anti-TB treatment were considered for the study.

2.2.2. Exclusion criteria

Patients with co-morbidities that may impact nutritional status, such as cancer, HIV/AIDS, or liver disease, were excluded. Additionally, pregnant or lactating women, individuals with severe mental illness or cognitive impairment, and those receiving nutritional support or therapy were not eligible for participation.

By applying these criteria, we aimed to select a population of TB patients without significant confounding factors, allowing for a comprehensive assessment of malnutrition risk using the MUST and PG-SGA tools. Participants were required to provide informed consent and have the capacity to complete the MUST and PG-SGA assessments.

2.3. Sample size

A total of 194 TB patients were enrolled in this study, which was conducted over a sixteen-month period. The sample size was determined based on the prevalence of malnutrition among TB patients, ensuring sufficient statistical power for meaningful analysis. Studies have reported malnutrition prevalence in TB patients at 52 % indicating undernutrition with an estimated 50 % prevalence used for sample size calculation.[9]

To determine the required sample size, we used the prevalence-based formula:

$$n = Z^2 \times p \times (1-p) / E^2$$

where:

$Z = 1.96$ (for a 95 % confidence level),

$p = 0.5$ (expected prevalence of malnutrition in TB patients),

$E = 0.0704$ (margin of error at ±7.04 %)

Substituting these values:

$n = (1.96)^2 0.5 \times (1-0.5)/(0.0704)^2$

$n = 193.93$

$n = 194.$

This ensured adequate statistical power to detect significant differences in patient characteristics and malnutrition risk while maintaining precision in prevalence estimation.

To ensure a representative sample, we enrolled all eligible TB patients who attended the clinic during the study period until the target sample size of 194 was reached. This approach helped minimise selection bias and enhance the generalizability of findings. Additionally, this sample size allowed for subgroup analyses, facilitating the exploration of potential associations between malnutrition risk factors and patient characteristics, thus improving the validity and reliability of our results.

2.4. Assessment tools

Malnutrition risk was assessed using two validated instruments named as MUST[6] and the PG-SGA.[7] These were chosen for their strong validation, universal use in clinical nutrition studies, and consistency with our study's aims to assess both rapid screening MUST and detailed assessment PG-SGA of malnutrition.

The MUST questionnaire, first introduced by Elia et al., scores malnutrition risk across three dimensions: BMI, unintentional weight loss, and acute disease influence, with the total score falling between 0 and 2 (≥1 score denoting increased risk). The PG-SGA, conceived by Ottery et al., combines patient self-report measures (e.g., weight history, dietary intake, symptoms) with clinical assessment tools (e.g., physical exam, functional status) to sort patients into categories of well-nourished, moderately malnourished, or severely malnourished.

Questionnaires were personally administered by the Principal Investigator (PI), who is bilingual in English and Hindi. Although the original English MUST and PG-SGA questionnaires were maintained to ensure their validated format, questions were explained in Hindi orally as administered, as the main language of the population under study was Hindi. This ensured that the participants understood the content to the fullest extent while ensuring the integrity of the standardized tools.

In order to make the questionnaires culturally and contextually sensitive, a multidisciplinary team of experts commented on the clarity and suitability of the questionnaires in the local healthcare setting. Small modifications were introduced to examples used in symptoms-related questions (e.g., replacing culturally known foods in eating questions) without modifying the fundamental scoring domains or criteria. A pilot with 20 participants was conducted to pretest the translated tools to ensure clarity, logical sequence, and ease of use. Simplification of the phrasing of two PG-SGA items was made based on feedback from the pilot to facilitate participant understanding. The bilingualism of the PI also ensured translation consistency in both pilot and main study phases.

BMI was calculated using measured height and weight (kg/m^2). In order to conduct further analyses, the BMI categories were established depending on the classification described by the World Health

Organization (WHO). These categories include underweight (BMI <18.5 kg/m^2), normal weight (BMI 18.5–24.9 kg/m^2), overweight (BMI 25.0–29.9 kg/m^2), and obese (BMI ≥30.0 kg/m^2).[10] All anthropometric data were collected by trained staff using standardised protocols to minimise measurement bias.

2.5. Data collection

Researchers gathered data using a structured questionnaire, which included demographic information, clinical data, and anthropometric measurements. The data was collected through a combination of interviews, questionnaires, and anthropometric measurements. The interviews focused on demographic information such as age, gender, occupation, and education level, as well as medical history such as duration of TB diagnosis, treatment regimen, and any other health conditions. Anthropometric measurements, including height, weight, and BMI, were obtained from patients' medical records. To ensure accuracy, anthropometric measurements were taken twice by a medical practitioner and one of the study's co-authors.

2.6. Statistical analyses

The statistical analysis was conducted using IBM SPSS Statistics for Windows, Version 27.0. Continuous variables such as age, height, weight, and BMI were analyzed using ANOVA with post hoc Bonferroni tests and independent samples t-tests to compare means across groups and between two groups, respectively. Non-normally distributed data and ordinal variables were represented by medians and interquartile ranges, while categorical data were presented as frequencies and percentages. To assess differences in patient characteristics and malnutrition risk across BMI groups, chi-square tests were used for categorical variables, and Kruskal-Wallis and Mann-Whitney U tests were used for ordinal variables. The agreement between malnutrition risk assessments made by MUST and PG-SGA was evaluated using Cohen's kappa, with interpretation based on Landis and Koch's criteria. A threshold of $p < 0.05$ was chosen to determine statistical significance.

2.7. Ethical considerations

This study was conducted in accordance with ethical principles to ensure the protection and privacy of participants. Written informed consent was obtained from all patients before participation, and their personal and medical information was kept confidential and de-identified. The study posed minimal risk to patients, and the potential benefits of the study outweighed the risks. As TB patients are vulnerable, extra precautions were taken to ensure their protection. The researchers were experienced in TB care and nutrition assessment, and the study received ethical approval from the Institutional Ethics Committee (RMRI/EC/13/2024). Data was stored securely and in accordance with relevant data protection regulations.

3. Results

3.1. Patient characteristics

The demographic and anthropometric characteristics of the 194 TB

Table 1
Demographic and anthropometric characteristics of patients.

Variables	Total n = 194	Underweight n = 145	Normal weight n = 46	Overweight n = 3	Obese n = 0	*p*-value
Age (years)	35.9 ± 17.3	35.2 ± 18.3	36.9 ± 13.8	54 ± 1.7	0	0.160
Gender						0.167
Male	118(60.8)	91(62.8)	24(52.2)	3(100)	0	
Female	76(39.2)	54(37.2)	22(47.8)	0(0)	0	
Height (cm)	160.2 ± 9.8	161.0 ± 10.0	157.7 ± 8.9	159.0 ± 1.7	0	0.131
Weight (kg)	42.2 ± 9.6	38.8 ± 7.4	51.6 ± 7.2	67.0 ± 1.7	0	<0.001*
BMI (kg/m^2)	16.3 ± 3.4	14.7 ± 1.9	20.6 ± 2.2	26.3 ± 0.1	0	<0.001*
Residence						0.011
Rural	171(88.1)	130(89.7)	40(87)	1(33.3)	0	
Urban	23(11.9)	15(10.3)	6(13)	2(66.7)	0	
Occupation						0.013
Daily Labour	106(54.7)	83(57.2)	22(47.8)	1(33.3)	0	
Farmer	14(7.2)	12(8.3)	2(4.3)	0(0)	0	
Fisherman	23(11.9)	20(13.8)	3(6.5)	0(0)	0	
Service employee	18(9.3)	11(7.6)	7(15.2)	0(0)	0	
House Wife	12 (6.2)	6(4.1)	6(13)	0(0)	0	
Jobless	1(0.5)	1(0.7)	0(0)	0(0)	0	
Merchant	9(4.6)	5(3.4)	3(6.5)	1(33.3)	0	
Student	11 (5.7)	7 (4.8)	3 (6.5)	1 (33.3)	0	
Family income						0.09
Poor	94(48.5)	69(47.6)	24(52.1)	1(33.3)	0	
Lower class	51(26.3)	45(31)	5(10.9)	1(33.3)	0	
Upper lower class	25(12.9)	17(11.7)	8(17.4)	0(0)	0	
Lower middle class	20(10.3)	12(8.3)	7(15.2)	1(33.3)	0	
Upper middle class	4(2.06)	2(1.4)	2(4.3)	0(0)	0	
Upper Class	0 (0)	0 (0)	0 (0)	0 (0)	0	
Education						0.410
Illiterate	89(45.9)	71(48.9)	16(34.8)	2(66.7)	0	
Primary	32(16.5)	20(13.8)	12(26.1)	0(0)	0	
Secondary	49(25.3)	37(25.5)	12(26.1)	0(0)	0	
Tertiary	24 (12.4)	17 (11.7)	6 (13)	1 (33.3)	0	
Marital Status						0.656
Divorced	2(1)	2(1.4)	0(0)	0(0)	0	
Married	109(56.2)	82(56.6)	26(56.5)	1(33.3)	0	
Single	83 (42.8)	61 (42.1)	20 (43.5)	2 (66.7)	0	

Significant *p* value considered at 0.05 level (two tailed), such p values marked as Asterisk (*) in superscript.
Continuous variables age, height, weight, and BMI were analyzed using ANOVA with post hoc Bonferroni tests and independent samples t-tests.
"Daily labour" refers to unskilled or semi-skilled workers who rely on daily wages for their livelihood. These individuals typically engage in manual labour-intensive work, such as construction, agriculture, sanitation work, or factory labour, often without long-term employment contracts.

patients are presented in Table 1. The majority of patients were male (60.8 %, n = 118), with a mean age of 35.9 years (±17.3). The mean weight of the patients was 42.2 kg (±9.6), with a statistically significant difference in mean weight ($p < 0.001$). The average BMI among all patients was 16.3 kg/m^2 with a standard deviation of 3.4. The BMI values for underweight, normal weight, and overweight groups were 14.7 kg/m^2, 20.6 kg/m^2, and 26.3 kg/m^2, respectively, with no participants classified as obese ($p < 0.001$). These findings indicate the possibility of a high prevalence of malnutrition risk among TB patients in this study population.

To compare differences in patient characteristics between BMI groups, one-way ANOVA was used to test continuous variables like age, height, weight, and BMI. Post hoc Bonferroni tests were then carried out for pair-wise comparisons to determine significant differences between BMI groups. To examine the relationship between BMI categorisation and malnutrition risk according to MUST and PG-SGA evaluations, chi-square tests were employed.

3.2. Prevalence of risk for malnutrition

There were notable variations in BMI classification among patients with underweight, normal weight, and overweight in relation to their malnutrition risk as evaluated by both the MUST and PG-SGA instruments ($p < 0.001$). Table 2 represents the evaluation of malnutrition risk among patients.

According to the MUST test, 74.2 % (n = 144) of all patients (n = 194) were categorised as being at a high risk for malnutrition (MUST score ≥2). The risk of malnutrition varied significantly between different BMI groups ($p < 0.001$). Within the underweight group (n = 145), 96.5 % (n = 140) were found to be at high risk. In the normal weight category (n = 46), 8.6 % (n = 4) were of high risk only, and none of the patients in the overweight category (n = 3) were of high risk for malnutrition. There were also 6.2 % (n = 12) of the whole population with a MUST score of 0 representing low risk, with a larger percentage in both the normal weight and overweight groups.

The PG-SGA evaluation also showed a very high rate of malnutrition risk among the patients. As many as 88.1 % (n = 171) of the patients presented with a PG-SGA value of ≥9, indicating a high risk of malnutrition. The rates were significantly different among BMI categories ($p < 0.001$). For the underweight patients, 88.3 % (n = 128) presented with high risk. Among the normal weight patients, 86.9 % (n = 40) were labelled as high risk, and among all overweight patients (100 %, n = 3), all of them were high risk for malnutrition. Obese patients were not present in this study.

With regard to international staging, patients were equally divided between Stage A (49.5 %), Stage B (30.9 %), and Stage C (19.6 %), with no correlation between BMI groups and severity of disease ($p = 0.665$). This indicates that malnutrition risk is not only a function of BMI but also of other factors, including decreased dietary intake, metabolic changes, and weight loss due to disease. The findings highlight the requirement for a double assessment approach since PG-SGA is found to give a broader picture of the risk of malnutrition among TB patients in contrast to MUST. Since MUST depends mainly on BMI and recent weight loss, and PG-SGA also takes dietary intake, symptom burden, and functional status into account, the results here imply that the integration of both measures may be essential for better malnutrition detection among TB patients.

3.3. Identifying malnutrition risk in underweight patients

Table 3 depicts the nutritional risk assessment using MUST and PG-SGA. According to MUST, while analysing various BMI groups (BMI <18.5), it was observed that 144 individuals from the underweight group had a higher risk for malnutrition compared to the overweight group ($p < 0.001$). Further analysis using all components of the MUST instrument showed that high-risk patients had a median score of 2, while low-risk patients scored 1 for BMI.

Out of a total of 194 patients, 145 underweight individuals with a BMI less than 18.5 were found to be at risk based on the PG-SGA score ($p < 0.001$). Significant differences were observed in all components of the PG-SGA scores ($p < 0.001$). There were noticeable differences in symptoms between high-risk and low-risk groups, with the former

Table 3
Nutritional risk assessment based on MUST & PG-SGA.

MUST	Median [range]			*p*-value
	Low risk	*Medium Risk*	*High risk*	
BMI score	1[0–2]	1[1–2]	2[0–2]	<0.001*
Weight loss score	2[0–2]	2[1–2]	2[2-2]	
Acute disease effect score	0[0-0]	0[0-0]	0[0-0]	
PG-SGA	**Median [range]**			***p*-value**
	Low risk	*Medium Risk*	*High risk*	
Symptoms	0[0-0]	3[0–5]	7 [0–12]	<0.001*
Food Intake	0[0-0]	2[0–2]	2[1–3]	
Activities & function	0[0-0]	0[0–1]	2[0–3]	
Worksheet 1 – Scoring Weight Loss	2[0–2]	2[1–2]	2[1–2]	
Worksheet 2 – Disease and its relation to nutritional requirements	1[1-1]	0[0–1]	1[0–2]	
Worksheet 3 – Metabolic Demand	0[0-0]	0[0–5]	0[0–6]	
Worksheet 4 – Physical Exam	0[0-0]	0[0-0]	0[0–2]	
Worksheet 5 – PG-SGA Global Assessment Categories	Median [range]			
	Stage A	Stage B	Stage C	
	1[1-1]	7[4–8]	13 [9–23]	

*Significant *p* value considered at 0.05 level (two tailed) such p values marked as Asterisk (*) in superscript.

Table 2
Assessment of malnutrition risk using MUST and PG-SGA.

Scoring categories	Total n = 194	Underweight n = 145	Normal weight n = 46	Overweight n = 3	Obese n = 0	*p*-value
MUST score						<0.001*
0	12(6.2)	2(1.4)	7(15.2)	3(100)	0	
1	38 (19.6)	3(2.1)	35(76.1)	0(0)	0	
≥ 2	144(74.2)	140(96.5)	4(8.6)	0(0)	0	
PG-SGA score						<0.001*
0–3	2(1)	0(0)	2(4.3)	0(0)	0	
4–8	21(10.8)	17(11.7)	4(8.6)	0(0)	0	
≥ 9	171 (88.1)	128(88.3)	40 (86.9)	3 (100)	0	
Global stage						0.665
A	96(49.5)	72(49.7)	22(47.8)	2(66.7)	0	
B	60(30.9)	43(29.7)	16(34.8)	1(33.3)	0	
C	38(19.6)	30(20.7)	8(17.4)	0(0)	0	

*Significant *p* value considered at 0.05 level (two tailed) such p values marked as Asterisk (*) in superscript.

exhibiting a median score of 7 compared to 0 in the latter. Similarly, there were differences in food intake, with the high-risk group scoring a median of 2 and the low-risk group scoring 0. Additionally, there were discernible divergences in daily activities and functioning, with the high-risk group scoring a median of 2, while the low-risk group scored 0.

3.4. Correlation of MUST and PG-SGA in identifying risk of malnutrition

The comparison between the MUST and PG-SGA scales based on BMI revealed minimal agreement ($\kappa = 0.010$, $p < 0.001$) between the two assessment tools (Table 4).

Among the underweight group consisting of 145 patients, 10.34 % were identified as high risk by the MUST assessment, while the PG-SGA evaluation categorised them as medium risk. Within the normal weight group (n = 46), 4.44 % were classified as low risk by MUST but as medium risk by PG-SGA, while 6.66 % were deemed high risk by PG-SGA and low risk by MUST. Notably, a significant 73.33 % were labelled high risk by PG-SGA but only medium risk by MUST. In the higher BMI subset (n = 3), all patients (100 %) were categorised as high risk by PG-SGA, but as low risk by MUST. These findings highlight disparities in the assessment of malnutrition risk across different BMI categories as determined by the two instruments.

4. Discussion

4.1. Key findings

This study highlights the significance of malnutrition risk among TB patients in Bihar, India, using both the MUST and the PG-SGA. Our findings reveal a high prevalence of malnutrition risk, with MUST detecting 74.2 % and PG-SGA detecting 88.1 % in the overall population. Notably, in patients with underweight, the prevalence of high malnutrition risk was even higher, with MUST identifying 96.5 % and PG-SGA identifying 88.2 %. This is concerning, as underweight status can exacerbate TB symptoms and impede recovery[11].

The differences in BMI classification between patients with low, normal, and overweight status, in comparison to their risk of malnutrition, indicate that BMI may not be an adequate measure to determine the risk of malnutrition in this group. The lack of obese patients in the study can be explained by the particular demographic features of the study population or the chronic disease status. The high risk of malnutrition among TB patients is a cause for concern, with almost three-quarters of the population being high-risk based on the MUST assessment. The close link between underweight and high malnutrition risk concurs with existing literature, demonstrating the susceptibility of underweight patients to nutritional deficiency[12–14].

Table 4
Cross-tabulation of risk assessment of malnutrition.

BMI classification	PG-SGA score category	MUST score category			
		0 Low Risk	*1 Medium risk*	*≥2 High risk*	Total
BMI <18.5 kg/m²	*0–3 Low risk*	0	0	0	0
	4–8 Medium risk	0	2	15	17
	≥9 High risk	2	1	125	128
	Total	2	3	140	145
BMI 18.5–24.9 kg/m²	*0–3 Low risk*	2	0	0	2
	4–8 Medium risk	2	0	2	4
	≥9 High risk	3	35	2	40
	Total	7	35	4	46
BMI ≥25.0 kg/m²	*0–3 Low risk*	0	0	0	0
	4–8 Medium risk	0	0	0	0
	≥9 High risk	3	0	0	3
	Total	3	0	0	3

Comparative evaluation of the MUST and PG-SGA scales indicates significant discordance in the determination of malnutrition risk in different BMI categories. The lowest value for concordance ($\kappa = 0.010$, $p < 0.001$) between the two tools implies that the tools cannot be used interchangeably to assess malnutrition risk in TB patients. PG-SGA identified malnutrition risk in 86.9 % of normal-weight and 100 % of overweight individual patients, who MUST be classified as high risk. This divergence emphasises the limitations of BMI-based screening and highlights the higher sensitivity of PG-SGA in detecting subclinical or non-apparent malnutrition. Unlike MUST, which relies primarily on BMI and weight loss, PG-SGA captures broader clinical dimensions, including symptom burden, dietary intake, functional status, and disease impact[6,7]. These attributes make PG-SGA a more comprehensive and clinically relevant tool for malnutrition assessment in TB. Its structure incorporates both clinician evaluation and patient-reported outcomes, offers graded risk staging, and has been validated across chronic diseases, such as cancer and HIV[15,16].

4.2. Clinical implications

The routine application of PG-SGA may facilitate personalised dietary planning, improve clinical management, and reduce disease-associated morbidity among TB patients. Unlike earlier studies, which primarily relied on BMI or single screening tools for nutritional assessment in TB patients, this study provides a comparative evaluation of both MUST and PG-SGA, revealing the limitations of BMI-based assessments and underscoring PG-SGA's broader diagnostic sensitivity. This dual-tool approach contributes to the existing TB-nutrition literature by providing a validated framework for identifying at-risk patients in low-resource settings.

Given the hypermetabolic state associated with tuberculosis, targeted nutritional interventions are essential[17]. A holistic nutritional assessment is necessary, incorporating dietary history, laboratory and clinical investigations, and functional measures, in addition to routine screening tools[18]. Malnutrition in TB patients is not only a secondary consequence but also a contributing factor to disease progression and treatment response[19]. Protein-rich diets providing 15–30 % of total energy intake are crucial for preserving muscle mass, supporting immune function, and promoting tissue repair[11]. However, dietary recommendations must be adapted to local food availability, cultural eating patterns, and socioeconomic limitations[20].

In resource-constrained settings, affordable protein-dense options such as legumes, milk, eggs, and fortified cereals can serve as practical alternatives[21]. Additionally, addressing deficiencies in key micronutrients like vitamins A, D, E, zinc, and iron through supplementation may be necessary[22]. Regular nutritional counselling and follow-up have been shown to improve cure rates and reduce relapse risk, highlighting the need to integrate comprehensive nutrition support into TB care protocols[23].

Public health systems, both in India and globally, should actively promote the integration of validated nutritional assessment tools such as PG-SGA into TB control programs. Incorporating PG-SGA within frameworks like the Revised National Tuberculosis Control Program (RNTCP) would facilitate the timely identification of malnourished patients, enabling prompt and tailored dietary interventions. Community health workers, especially those involved in directly observed treatment short-course (DOTS) implementation, can be trained to administer PG-SGA during routine visits. This integration would foster early detection of nutritional risk, enhance treatment adherence, and ultimately improve clinical outcomes in TB patients, particularly in low-resource, high-burden settings. Although this study focused on a regional population in India, its implications extend to the National Tuberculosis Elimination Program (NTEP) and similar high-burden TB settings.

4.3. Limitations

This research has several potential limitations, including a small sample size and reliance on self-reported food intake data. The absence of biochemical markers of nutrition, such as albumin, total protein concentrations and iron levels, restricts a more detailed assessment of nutrition. Future longitudinal follow-up studies should evaluate the impact of nutritional interventions on success rates and investigate biochemical and functional indices of malnutrition beyond anthropometric assessments.

Despite these limitations, this research provides valuable insights into the differences between MUST and PG-SGA assessments in TB patients, highlighting the importance of multimodal screening for malnutrition risk. By using structured nutritional evaluations and individualised dietary interventions, healthcare professionals can develop tailored nutrition plans that address specific patient vulnerabilities, ultimately reducing morbidity and mortality rates in TB patients. Future research should focus on combining multiple screening tests with objective biochemical indicators to establish a gold-standard approach for assessing malnutrition in TB patients.

5. Conclusion

This study concludes that malnutrition remains a significant concern among TB patients, with high prevalence of 74.2 % and 88.1 % as identified by MUST and PG-SGA, respectively. The findings highlight that underweight patients are disproportionately affected, with both screening tools identifying a high percentage of patients at risk. This highlights the importance of early detection and intervention in preventing malnutrition-related complications in TB care. The observed disagreement between MUST and PG-SGA risk categorisation suggests that a comprehensive approach, incorporating clinical, dietary, and functional assessment, is critical for accurate malnutrition screening. These findings support the incorporation of PG-SGA as a routine screening tool within national TB control programs such as the NTEP, enabling early detection of malnutrition and targeted nutritional support at the community level. This study supports the WHO End TB Strategy's emphasis on integrated, patient-centred care by advocating for routine nutritional screening and intervention as a part of comprehensive TB management. By identifying and addressing malnutrition using validated tools like PG-SGA and MUST, this study promotes a shift toward holistic, person-centred care. Furthermore, the findings align with the Sustainable Development Goals, particularly SDG 2 (Zero Hunger), SDG 3 (Good Health and Well-being), by targeting undernutrition and enhancing treatment outcomes. Integrating nutritional screening and care into national TB control strategies is essential for improving patient outcomes and advancing progress toward global TB elimination.

Ethical consideration

The study received ethical approval from the Institutional Ethics Committee (RMRI/EC/13/2024). Data was stored securely and in accordance with relevant data protection regulations.

Contribution

Conceptualization was done by P.G., K.*P. and* S.D. The study design was developed by P.G., K.*P. and* S.D. Data curation was done by P.G., M. A*., M.A**, R.K.M., T.K., and G.S.R. Data analysis and interpretation were conducted by P.G., A.S., K.*P. and* S.D. The manuscript was drafted by P.G. Critical revision of the manuscript was performed by A.S., K. *P. and* S.D. Supervision of the study was provided by K.*P. and* S.D. All authors reviewed and approved the final version of the manuscript.

M.A* M.A** refer to Meenakshi Abbagouni, Maturu Ashok respectively.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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